Lenses and image formation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use ray directions to explain converging and diverging lens behavior.

SUCCESS CRITERIA - I can...

1. Distinguish converging and diverging lenses.
2. Trace principal rays with a clear optical axis.
3. State whether an image is real or virtual in the given case.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “lens” and “focal point” to explain your model or result. Sentence starters are optional.

lens: a transparent element that changes light-ray directions

focal point: where suitable parallel rays meet or appear to originate

ray: a line showing the modeled direction light travels

01 Warm-up: bring back an earlier idea

A source approaching a stationary listener at 15 m/s emits 500 Hz.
Use $v_{sound}=340$ and $f_{obs}=f_s\ v/(v-v_s)$.
Find observed frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

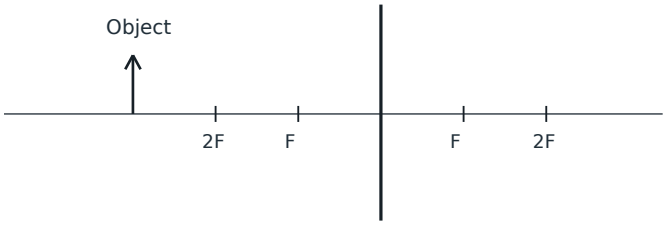
02 Worked example: explain the model

Reference: $1/f = 1/d_o + 1/d_i$.
Thin lens: f is focal length; o = object, i = image. Use the stated sign convention and a ray diagram.

Worked reasoning: A converging lens can form a real image of an object beyond its focal point. A ray parallel to the axis passes through the far focus; a central ray is approximately undeviated in the thin-lens model.

Your turn: Draw these two rays and mark their intersection for an object beyond 2 f.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Complete ray sketches for parallel rays through a converging lens and a diverging lens.
Label the optical axis 0.

Sentence starter: The … in my model represents … . Its label shows … .



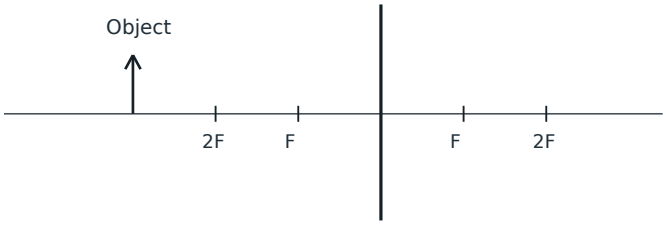
Left: converging lens. Right: diverging lens. Horizontal lines mark the optical axes.

Marzano check - circle: ① ② ③

04 Apply the physics

A converging lens has focal length 10 cm. A small lamp is 30 cm from it.
Label the model, find image distance using $1/f = 1/d_o + 1/d_i$, and calculate $m = -d_i/d_o$.
Check the sign and size with two principal rays.

Sentence starter: I found … [unit]. This value means … in this situation.



Converging thin lens; add principal rays.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A magnifying glass produces an upright virtual image when object is inside f.
Explain why “converging lens always makes a real image” is wrong.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A wave snapshot has crests at 0.2, 0.6, 1 m.
Find wavelength. Its period is 0.06 s; find frequency.
Label a wave sketch to support the calculation or comparison.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.



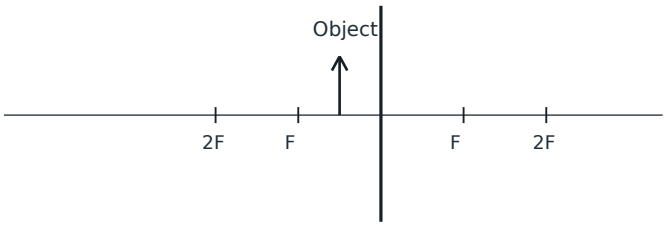
Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

For a converging lens f=10 cm and d_o=5 cm, find d_i and m.
State image kind.

Sentence starter: I found ... [unit]. This value means ... in this situation.



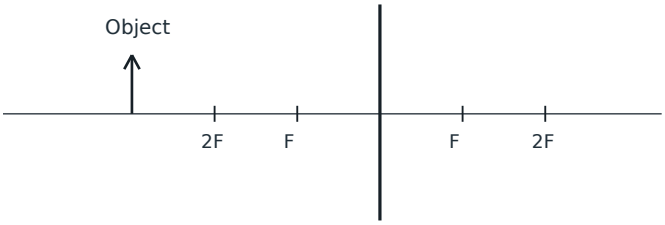
Converging thin lens; add principal rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A diverging lens has f = -10 cm and an object 30 cm in front.
Use $1/f = 1/d_o + 1/d_i$ and $m = -d_i/d_o$.
Find image distance and magnification, then check them with a ray sketch.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Diverging thin lens; add principal rays.

Marzano check - circle: 1 2 3